

Astrid Arena Olave Herrera

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EDUCATION

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| <p>Ph.D. in Mathematics
 <i>Michigan State University</i>
 Advisor: Elizabeth Munch</p> | <p>2021 - Present</p> |
| <p>M.Sc. in Applied Mathematics
 <i>Universidad Nacional de Colombia</i>
 Meritorious thesis : Revealing brain network dynamics during the emotional state of suspense using topological data analysis
 Advisor: José Perea (MSU). Coadvisor: Francisco Gómez</p> | <p>2021</p> |
| <p>B.Sc. with Honors in Mathematics
 <i>Universidad Nacional de Colombia</i>
 Thesis: Persistent homology analysis of RNA secondary structures
 Advisor: Gustavo N. Rubiano</p> | <p>2018</p> |

RESEARCH INTERESTS

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| <ul style="list-style-type: none"> • Topological data analysis:
Foundations and applications | <ul style="list-style-type: none"> • Algebraic topology • Mathematical biology |
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PAPERS

4. Astrid A. Olave. A new family of distances over partially ordered sets. *ArXiv e-prints*, June 2026, 2606.06377
3. Astrid A. Olave and Elizabeth Munch. Bounding the interleaving distance on concrete categories using a loss function. *ArXiv e-prints*, January 2026, 2601.09034
2. Astrid A. Olave, Jose A. Perea, and Francisco Gómez. Revealing brain network dynamics during the emotional state of suspense using tda. *Network Neuroscience*, page 1–25, August 2025. doi:10.1162/netn.a.34

1. Konstantinos Georgiou, Andy (Jia) Jiang, Edward Lee, Astrid A. Olave, Ian Seong, and Twesh Upadhyaya. Lift & project systems performing on the partial-vertex-cover polytope. *Theoretical Computer Science*, 820:1–16, June 2020. doi:10.1016/j.tcs.2020.03.004

PRESENTATIONS

16. *Bounding the interleaving distance on concrete categories: Toward polynomial time algorithms.* Talk in the Joint Mathematics Meetings 2027. McCormick Place , IL, US. Jan 2027
15. *From climbing to diving: defining a new family of poset metrics.* Talk in Combinatorics and Graph Theory Seminar. Michigan State University, MI, US. Dic 2026
14. *Bounding the interleaving distance between multiparameter persistence modules in polynomial time.* Talk in the 2026 AMS Fall Western Sectional Meeting. Arizona State University, AZ, US. Nov 2026
13. *From climbing to diving: defining a new family of poset metrics.* Talk in undergraduate colloquium. Kalamazoo College, MI, US. Oct 2026
12. *Bounding the interleaving distance on concrete categories: Toward polynomial time algorithms.* Talk in the 2026 AMS Fall Eastern Sectional Meeting. George Washington University, DC, US. Oct 2026
11. *The topology of suspense: Using TDA to reveal brain network dynamics.* Talk in the Student Applied Mathematics seminar. Michigan State University, MI, US. Sep 2025
10. *Bounding the interleaving distance on concrete categories using a loss function.* Talk in the Applied Algebraic Topology Research Network (AATRN). Online Seminar. Aug 2025
9. *Bounding the interleaving distance on concrete categories using a loss function.* Poster present at the workshop: The geometric realization of AATRN. Institute for Mathematical and Statistical Innovation (IMSI), IL, US. Aug 2025
8. *Bounding the interleaving distance on concrete categories using a loss function.* Talk in the SIAM Conference on Applied Algebraic Geometry (AG25). University of Wisconsin-Madison, WI, US. Jul 2025
7. *The history of the interleaving distance: A journey from homology to its applications.* Talk in colloquium. Universidad de Málaga, España. Jun 2025
6. *Generalizing the interleaving distance using categorical flows.* Short video on the AATRN tuthorial-a-thon Spring 2025. YouTube video. Feb 2025
5. *The history of the interleaving distance: A journey from homology in $\mathbb{R}^3$ to categorical flows.* Talk in the Student Geometry and Topology seminar. Michigan State University, MI, US. Oct 2024

4. *Revealing brain network dynamics during the emotional state of suspense using topological data analysis.* Oral contribution presented at GEOTOP-A International Conference: Applications of Geometry and Topology. Universidad Autónoma de Yucatán, México. Jan 2024
3. *Finding interleaving distances on a poset using categories with flows.* Poster present at 2023 AWM Research Symposium. Atlanta, GA, US. Oct 2023
2. *Revealing brain network dynamics during the emotional state of suspense using topological data analysis.* Short talk presented at II SIPAIM virtual Workshop. Online Seminar. May 2021
1. *Persistent homology analysis of RNA secondary structures.* Poster session presented at XXII Colombian Congress of Mathematics. Universidad del Cauca, Colombia. Jun 2019

HONORS AND AWARDS

GTA Award for Excellence in Teaching <i>Michigan State University</i>	2025
Dr. Paul and Wilma Dressel Endowed Scholarship <i>Michigan State University</i>	2025
Mathematics Recruiting Fellowship <i>Michigan State University</i>	2021
Scholarship for complete tuition for M.Sc programs <i>Universidad Nacional de Colombia</i>	2019

TEACHING EXPERIENCE

Michigan State University

Matrix Algebra with Computational Applications (MTH 314) <i>Teaching assistant of section 004, Instructor of section 101</i>	Fall 2024, Summer 2023
Introduction to proofs (MTH 299) <i>Teaching assistant of sections 001 and 003</i>	Spring 2026, Fall 2025
Calculus III (MTH 234)	Spring 2024

Teaching assistant of sections 001,015 and 018

Calculus II (MTH 133)

Spring 2023

Teaching assistant of sections 013,018 and 024

Calculus I (MTH 132)

Fall 2022

Teaching assistant of sections 019,021 and 024

Survey of Calculus I (MTH 124)

Fall 2021

Teaching assistant of Lecture 1

College Algebra II (MTH 103B)

Spring 2025, Fall 2024

Instructor of section 010, 001

Quantitative literacy (MTH 101)

Summer 2026

Teaching assistant of section 102

TEACHING ADJACENT EXPERIENCE

Orientation Leader for First-Year PhD Students

2025 - 2026

Michigan State University

Organized monthly meetings to help incoming PhD students transition into the teaching and academic environment of MSU.

Leader for teaching assistants in MTH 234

Spring 2024

Michigan State University

Organized biweekly meetings to support teaching assistants and discuss lesson preparation

Conduct classroom observations

Fall 2023

Michigan State University

Provided teaching feedback for 8 teaching assistants

Mathematics Learning Center

Spring 2022

Michigan State University

Tutor

Integral calculus

First semester 2018

Universidad Nacional de Colombia

Writer of solutions of the book "Cálculo integral en una variable"

Mentorship

Summer Topology Program

Summer 2026

Michigan State University

Guide undergraduate students study TDA.

Tutor of Círculos Matemáticos

Second semester 2019

Universidad Nacional de Colombia

Guided enthusiastic high school students about mathematics into the experience of being a mathematician

PROFESSIONAL CERTIFICATES

Certification in College Teaching

2024 - 2025

Michigan State University

Project: The implementation Khanmigo as a mathematics tutor:
A study from students' perspectives

Data Science Boot Camp

May - June 2024

The Erdős Institute

Project: Temporal Graphs for Music recommendation systems

SERVICE

Departmental service

AMS Graduate Student Chapter at MSU

2024-2025

Treasurer

Council of Graduate Students

2023-2024

Representative of the mathematics department

Professional service

Topological Data Analysis Seminar

2025-2026

Organizer

Graduate Student Top. and Geo. Conference

April 2024

Organizer

Community outreach

Marble Elementary School Math Night

November 2024

Volunteer

Reviewer/Referee

EXTRACURRICULAR ACTIVITIES

Intramural soccer <i>Player and captain of the mathematics team</i>	2021-2025
Sin Medida inter-university catholic community <i>Leadership team member</i>	2018-2021
Universidad Nacional women's soccer team <i>Goalkeeper</i>	2014-2021